# GLP-1 / GLP-1R alanine-scanning mutagenesis

Literature data handout · Part 1: alanine scanning of the GLP-1 peptide

Primary source: Adelhorst K et al. (1994), Journal of Biological Chemistry 269:6275–6278. Table I and text. DOI: 10.1016/S0021-9258(17)37366-0.

Experiment: successive residue substitutions in GLP-1(7–36)-amide; naturally occurring alanines were replaced by the corresponding glucagon residues when possible. Binding measured at recombinant rat GLP-1R (CHL membranes); adenylyl cyclase activity in RIN 2A18 membranes.

## Wild-type GLP-1(7–36)-$NH_2$ sequence (proglucagon residue numbering):

7:H 8:A 9:E 10:G 11:T 12:F 13:T 14:S 15:D 16:V 17:S 18:S 19:Y 20:L 21:E

22:G 23:Q 24:A 25:A 26:K 27:E 28:F 29:I 30:A 31:W 32:L 33:V 34:K 35:G 36:R

## Selected experimentally reported binding / signaling results

| GLP-1 variant | Binding ($IC_{50}$ or relative loss) | Adenylyl cyclase / interpretation |
|---|---|---|
| WT GLP-1 | $IC_{50}$ = 0.27 ± 0.09 nM | $EC_{50}$ = 2.6 ± 0.4 nM |
| H7A | ~30 nM (~111× higher $IC_{50}$) | No measurable activation at tested concentrations up to 10 µM |
| G10A | Marked loss of binding affinity | No measurable activation up to 10 µM |
| F12A, T13A | Marked loss of affinity | N-terminal side chains implicated in receptor recognition |
| D15A | Marked loss of affinity | No measurable adenylyl cyclase activation up to 10 µM |
| S17A | $IC_{50}$ = 0.46 nM | Minor change in binding compared with WT |
| A24R | $IC_{50}$ = 0.89 nM | Glucagon-residue substitution at a native alanine position |
| F28A | $IC_{50}$ = 531 nM (~1,970× WT) | Substantial change in secondary structure observed by circular dichroism |
| I29A | Marked loss of affinity | Secondary-structure change; interpret as a conformational effect |

Interpretation: H7, G10, F12, T13 and D15 are the principal N-terminal sensitivity hotspots. F28 and I29 strongly affect the bound-compatible peptide conformation. In a later review, the five N-terminal substitutions were summarized as causing ~41–219-fold affinity losses in the original scan; this range should not be applied to each individual variant.

Scope note: These are selected, traceable values and qualitative findings, not a transcription of every numerical cell in Table I. The full alanine-substitution panel and exact per-variant $IC_{50}$/$EC_{50}$ results are in Adelhorst et al., Table I. Native A25 was not scanned in that series.

Supporting review: The structure and function of the GLP-1 receptor and its ligands, section on GLP-1 analog mutagenesis.

# GLP-1R extracellular-loop-2 alanine scan

Part 2: receptor mutagenesis, separated by binding and functional readouts

Primary source: Koole C et al. (2012), Journal of Biological Chemistry 287:3642–3658, Tables 1–4. DOI: 10.1074/jbc.M111.309328 · Open full text.

Experiment: 23 single alanine replacements over human GLP-1R positions 285–307 (ECL2 and adjacent residues), evaluated for cell-surface receptor expression, GLP-1(7–36)-$NH_2$ binding and cAMP, $Ca^{2+}$ and ERK1/2 signaling.

## Observed GLP-1(7–36)-$NH_2$ binding changes

| Binding category vs WT | Receptor variants | Interpretive notes |
|---|---|---|
| Significantly lower peptide affinity | K288A, E292A, D293A, C296A, W297A, R299A, N300A, N302A, M303A, Y305A, L307A | Several also decrease functional receptor binding or signal coupling; C296A and W297A have surface-expression / assay-window concerns. |
| No significant affinity change | G285A, I286A, V287A, Y289A, L290A, Y291A, E294A, G295A, T298A, S301A, N304A | Unchanged affinity does not establish normal potency or efficacy. |
| Binding not measurable | W306A | No detectable surface expression in this study; cannot infer a direct peptide contact from loss of response. |

## Selected signaling phenotypes (GLP-1 peptide stimulation)

| Receptor mutation | Functional observation / reading |
|---|---|
| K288A, E292A, C296A, W297A, N300A | Marked reduction in cAMP signaling; GLP-1(7–36)-mediated intracellular $Ca^{2+}$ response abolished. |
| R299A, Y305A | Relative bias toward pERK1/2 signaling compared with the WT receptor. |
| N302A, L307A | Decreased affinity / cAMP potency and increased relative pERK1/2 bias. |
| N304A | Binding affinity not significantly altered, but particularly large loss of cAMP coupling efficacy. |
| V287A vs T298A | V287A: decreased cAMP potency; T298A: increased potency despite reduced coupling efficacy. |
| W306A | No detectable cell-surface receptor or biological response in this assay system. |

Data interpretation: Affinity ($K_i$ / $pIC_{50}$), signaling potency ($EC_{50}$ / $pEC_{50}$), maximum response (Emax), expression-corrected efficacy (log τc), and pathway bias answer different questions. The Koole study normalizes functional efficacy for receptor surface expression. Mutations can change protein folding or trafficking as well as the peptide interface.

Optional independent receptor dataset: Dods R & Donnelly D. (2016), Bioscience Reports 36:e00285, DOI: 10.1042/BSR20150253. Its numerical GLP-1 $pEC_{50}$ / $pIC_{50}$ table includes K288A, R299A, N300A, Y305A and W306A. Assay systems differ; do not pool numeric estimates directly.